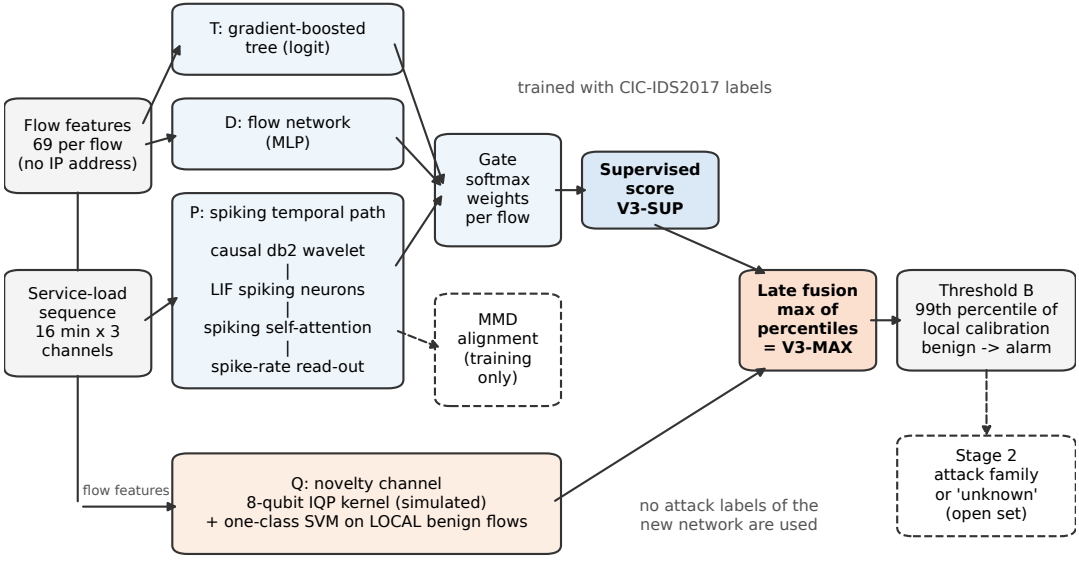
T: gradient-boosted tree (logit)
Flow features 69 per flow (no IP address)
D: flow network (MLP)
trained with CIC-IDS2017 labels
Gate softmax weights per flow
Supervised score V3-SUP
P: spiking temporal path
causal db2 wavelet
LIF spiking neurons
spiking self-attention
spike-rate read-out
Service-load sequence 16 min x 3 channels
MMD alignment (training only)
Late fusion max of percentiles = V3-MAX
Threshold B 99th percentile of local calibration benign -> alarm
Stage 2 attack family or 'unknown' (open set)
flow features
Q: novelty channel 8-qubit IQP kernel (simulated) + one-class SVM on LOCAL benign flows
no attack labels of the new network are used